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Adaptive Toeplitz convolution- enhanced classifier for anomaly detection in ECG big data

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Abstract

The anomaly detection of electrocardiogram (ECG) data is crucial for identifying deviations from normal heart rhythm patterns and providing timely interventions for high-risk patients. Various autoencoder (AE) models within machine learning (ML) have been proposed for this task. However, these models often do not explicitly consider the specific patterns in ECG time series, thereby impacting their learning efficiency. In contrast, we adopt a method based on prior knowledge of ECG time series shapes, employing multi-stage preprocessing, adaptive convolution kernels, and Toeplitz matrices to replace the encoding part of the AE. This approach combines inherent ECG features with the symmetry of Toeplitz matrices, effectively extracting features from ECG signals and reducing dimensionality. Our model consistently outperforms state-of-the-art models in anomaly detection, achieving an overall accuracy exceeding 99.6%, with Precision and Area Under the Receiver Operating Characteristic Curve (AUC) reaching 99.8%, and Recall peaking at 99.9%. Moreover, the runtime is significantly reduced. These results demonstrate that our technique effectively detects anomalies through automatic feature extraction and enhances detection performance on the ECG5000 dataset, a benchmark collection of heartbeat signals.

Keywords: ECG time series, Toeplitz convolution, Classifier, Anomaly detection

Introduction

The task of anomaly detection in big data appears across various domains of scientific and technological endeavors. For instance, in fields such as the Internet of Things, medical diagnostics, and industrial monitoring [1–3], anomaly detection in electrocardiogram (ECG) time series data stands out as a critical application [4]. The typical workflow of time series anomaly detection generally involves the following steps [5]: (1) partitioning the available time series dataset into training and testing sets; (2) selecting appropriate feature extraction methods; (3) establishing anomaly detection models to identify abnormal patterns; (4) conducting anomaly detection and evaluation on the testing set.

Conventional changes in ECG occur in various cardiovascular abnormalities, including atrial fibrillation, ventricular tachycardia, myocardial ischemia, localized myocardial necrosis, and electrolyte imbalances [6]. ECG signal analysis has long been utilized for diagnosing cardiac issues and remains a primary tool in preventing cardiovascular diseases [7]. However, challenges and areas for improvement persist in

ECG time series analysis: 1. Noise interference poses a significant issue, where electrocardiographic signals can be affected by various physiological and environmental factors [8]. 2. Adaptive feature space design remains problematic. Existing feature extraction methods often inadequately account for the shape, trends, and periodicity analysis of ECG time series, with optimal feature subsets varying across datasets [9]. 3. Selection strategies for anomaly detection models. Current methods either lean towards black-box models like deep learning-based approaches or rely on expert selection and hyperparameter tuning, limiting their generalizability and interpretability on new datasets [10]. 4. Temporal dependencies pose challenges despite mature solutions in automated machine learning for structured data [11].

This paper proposes a methodology based on prior knowledge of ECG time series shapes, designing an adaptive kernel extraction and constructing a Toeplitz convolution ECG-ATC model. Recognizing the importance of preceding context in predicting future events in time series data tasks, the convolutional pooling-enhanced classifier addresses potential pitfalls traditional methods may encounter when handling high-dimensional data, such as the curse of dimensionality [12]. The Toeplitz convolution integrates symmetric features of adaptive kernel functions and Toeplitz matrices, enhancing generalizability. While recurrent neural networks (RNNs) and long short-term memory networks (LSTMs) may struggle with long-term dependency, the Toeplitz convolution with adaptive kernels mitigates this to some extent, flexibly capturing local features within sequences. This approach improves handling of imbalanced datasets of normal and anomalous ECG signals, thereby enhancing anomaly detection capabilities.

The main contributions of this paper can be summarized as follows:

- Employing multi-stage signal enhancement methods in ECG time series preprocessing using various filtering and denoising techniques to improve signal quality and accuracy.
- Adaptive definition of convolution kernels for each ECG time series, facilitating personalized convolution operations to better capture individual sequence features and patterns, thereby enhancing model flexibility and adaptability across diverse sequences.
- Leveraging the computational efficiency and structural simplicity of Toeplitz matrix convolution operations, the proposed adaptive Toeplitz convolution ECG time series processing model (ECG-ATC) reduces computational complexity, accelerates convolution operations, and conserves computational resources.
- Evaluating the proposed processing approach on publicly available ECG datasets through classifier detection, validating its role in ECG time series anomaly detection.

Paper Organization: The rest of this paper is organized as follows. Section Related Research reviews the related work. Section Materials and Methods introduces the basic concepts and formally defines the ECG-ATC model and framework. Section Experimental Evaluation presents the experimental results of evaluating the preprocessing and anomaly detection. Section Experimental observations on data efficiency

of ECG-ATC methods provides the key observations we draw from the experimental study on the relationship between the model performance and the number of training samples. Section Conclusion summarizes this work and points out possible future directions.

Related research

In the realm of research on abnormal ECG detection, studies typically integrate machine learning approaches, including traditional supervised methods such as Support Vector Machines (SVM) and Random Forests [13, 14], as well as unsupervised techniques like clustering methods and anomaly detection algorithms [12]. Deep learning methods such as RNNs, LSTMs, and Transformers [15–17] are also employed. Waveform analysis techniques, such as QRS complex detection and shape analysis, are utilized for identifying anomalies or abnormal patterns [18, 19]. Ensemble learning, combining multiple techniques and models, aims to enhance the accuracy and robustness of anomaly detection [20].

To elevate the classification proficiency of arrhythmias, a novel end-to-end feature learning and classification model has been proposed [13]. This model employs an SVM classifier based on an autoencoder convolutional network (ACN) to detect four types of arrhythmic beats, including normal beats, using the MIT-BIH Arrhythmia Database. The model's statistical performance is evaluated using a tenfold cross-validation strategy, achieving an overall accuracy of 98.84%, average accuracy of 99.53%, sensitivity of 98.24%, and precision of 97.58%.

Myrovali et al. [14] describe a method utilizing ECG recordings during sinus rhythm to identify patients with a history of Paroxysmal Atrial Fibrillation (PAF). They identified the most discriminative features for classification and trained a Random Forest achieving 93.45% accuracy, 95.21% sensitivity, 91.40% specificity, and an AUC of 0.9669, offering potential value for early detection of PAF-prone patients.

Sonawane and Patil [12] applied a weighted feature extraction method combined with mixed clustering models (overlaying DBCAN on K-Means clustering) optimized with an improved variant of Chicken Swarm Optimization (MU-CSO), addressing the curse of dimensionality using Principal Component Analysis (PCA).

Deep learning has demonstrated remarkable capabilities in addressing various medical challenges. Prabhakararao and Dandapat [15] introduced a hierarchical structure-based Multi-level Multi-task Attention Recurrent Neural Network (MMA-RNN) employing Bi-directional LSTM and attention layers to discern hierarchical sequential features. Their approach enhances information synergy and reduces error accumulation through multi-head classifiers, achieving more concise and accurate analyses by incorporating clinical features into 12-lead ECG characteristics.

Sun et al. [16] proposed a weighted-shared recurrent neural network with intra- and inter-lead attention modules, mimicking clinical diagnostic procedures to derive representative vectors of attention. Their method achieved an improved overall accuracy of 98.3% on the PTB database. Transformer models are increasingly employed for temporal sequence modeling, with Akan et al. [17] proposing the ECGformer model for classifying various arrhythmias in ECG data, evaluated on MIT-BIH and PTB datasets.

ECG signals, composed of components like P waves, QRS complexes, and T waves, play a crucial role in cardiovascular disease management. Yuen et al. [18] a machine learning pipeline for detecting QRS complexes in highly noisy wearable ECG devices, employing ConvLSTM for superior performance compared to state-of-the-art algorithms in terms of F1 score.

Mandala et al. [20] proposed an enhanced ensemble learning approach, the Fine-Tuned Boosting Optimization (FTBO) model, for PVC, PAC, and AF arrhythmias in the MIT-BIH dataset, demonstrating high sensitivity, specificity, and accuracy across all three types of arrhythmias.

This study utilizes adaptive kernel function-based Toeplitz convolution to learn feature representations from unlabeled data, reducing dependence on labeled data and extending model applicability. Additionally, Toeplitz convolution effectively captures local structures of time-series data, enhancing classifier detection accuracy and robustness.

Materials and methods

ECG time series data

In the realm of ECG signal processing, a time series constitutes an organized collection of data points arranged in chronological order. Sampling, within this context, denotes the process of converting continuous signals into discrete digital signals. The sampling rate governs the number of samples collected per unit of time, ensuring their uniform distribution and forming a discrete sequence. Our work involves validation of the proposed model using the ECG5000 dataset [21], a widely adopted repository in the domain of electrocardiography for the detection of cardiac anomalies. This dataset comprises 5000 single-lead ECG signals, each of length 140 time points, commonly employed for the development and performance evaluation of heartbeat classification algorithms. This aids healthcare professionals in enhancing the diagnosis of cardiac ailments [22]. Each sample in this dataset is annotated with a corresponding cardiac arrhythmia category, encompassing normal heart rhythms and four types of arrhythmias: Normal, Premature Ventricular Contraction (PVC), R-on-T Premature Ventricular Contraction (R-on-T), Supraventricular Premature or Ectopic Beat (SP), and Unclassified Beat (UB) (Fig. 1).

The ECG5000 dataset consists of raw electrocardiogram signals without prior pre-processing. When utilizing this dataset for classification tasks, it is imperative to

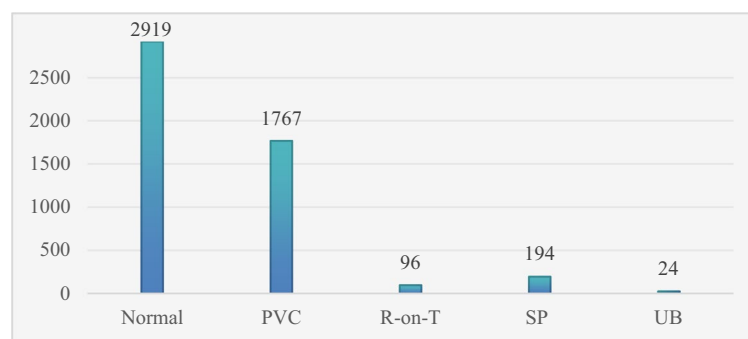


Fig. 1 The categorization distribution of the ECG5000 time series data

conduct preliminary preprocessing steps to eliminate interference and prepare the time series signals for detection purposes [23].

Data preprocessing

During the preprocessing of ECG time-series data, a multi-stage signal enhancement method known as composite filtering, denoising, and baseline removal (FDL) is employed. This approach incorporates various filtering and denoising techniques to enhance the quality and accuracy of the signals.

Time series data preprocessing

The multi-stage composite signal enhancement method, as proposed by FDL, involves the following computational processes for each ECG time series sample $\mathbf{x}_i = \{x_{i1}, \dots, x_{it}, \dots, x_{is}\}$, where s denotes the length of the sampling points. Initially, a composite filtering approach is applied to obtain the filtered signal value at time t . This composite filter integrates Notch Filtering and median filtering techniques [24], employing a specified window size of $2k + 1$ for the median filtering process. Among them, the transfer function of the notch filter can be expressed as:

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}} \quad (1)$$

The coefficients $\mathbf{b} = \{b_m, m = 1, 2, \dots, M\}$ and $\mathbf{a} = \{a_n, n = 1, 2, \dots, N\}$ refer to the Infinite Impulse Response (IIR) notch filter [25], where b_m represents the numerator coefficients of the filter at time instant m , and a_n represents the denominator coefficients of the filter at time instant n . M indicates the length of the numerator coefficients, and N indicates the length of the denominator coefficients.

In discrete time signal processing, median filtering can be implemented by sorting the values within a window of the signal and selecting the middle value. For each sample, the median filter can be expressed as:

$$\bar{x}_{it} = \text{median}\{x_{i,t-k}, x_{i,t-k+1}, \dots, x_{i,t+k}\} \quad (2)$$

$$\hat{x}_{it} = \frac{1}{2} \left[\bar{x}_{it} + \sum_{m=0}^M b_m (x_{i,t+k-m} + x_{i,t-k-m}) - \sum_{n=1}^N a_n (x_{i,t+k-n} + x_{i,t-k-n}) \right] \quad (3)$$

$\bar{x}_{it}$ is the median filtering result at time t , $x_{i,t+k-m}$ and $x_{i,t-k-m}$ are the values of the input signal at time $t + k - m$ and $t - k - m$. For a given cutoff frequency f_0 and quality factor Q , along with the sampling frequency f_s , the coefficients $b[m]$ and $a[n]$ of the filter can be determined using the Butterworth notch filter design method. The lengths M and N depend on the chosen filter order. Typically, Butterworth notch filters are second-order filters, thus M and N can both be 3.

Due to the non-stationary nature of ECG time series data, capturing local temporal features is essential [26]. Consequently, the second phase involves denoising the signal using wavelet transformation. Wavelet transformation is particularly advantageous in signal analysis and processing scenarios that require consideration of signal locality and multiscale analysis, especially in denoising applications [27]. In contrast, Fourier

transformation is more suitable for extracting fixed frequency domains and does not correspond to specific time points, providing only global information [28]. By employing specified wavelet basis functions and thresholds, the signal undergoes wavelet transformation, followed by soft-thresholding of wavelet coefficients, and finally, inverse wavelet transformation to reconstruct the denoised signal. This approach not only captures frequency information but also accurately pinpoints temporal locations. ECG time series data is a discrete signal [29], and wavelet series expansion is performed:

$$x_{it} = \sum_{j=1}^{\infty} \sum_{\tau \in \mathbb{Z}} c_{j,\tau} \psi_{j,k}(t) + \sum_{\tau \in \mathbb{Z}} d_{\tau} \phi_{j_0,k}(t) \quad (4)$$

where $\psi_{j,k}(t) = 2^{j/2} \psi(2^{j/2}t - \tau)$ is the wavelet mother function and $\phi_{j_0,k}(t) = 2^{j_0/2} \phi(2^{j_0}t - \tau)$ is the wavelet father function. $c_{j,\tau}$ and d_{τ} are coefficients corresponding to the father wavelet and mother wavelet, respectively. j is the scale factor, corresponding to the inverse of ω in the Fourier transform (period T). τ is the translation factor. By adjusting the scale and translation parameters, localized analysis of the signal in the time–frequency domain can be achieved.

The third stage involves removing the baseline drift component from the signal [30]. This entails averaging each time series sample, subtracting this average from each data point across the entire signal, thereby producing a new signal where the baseline (constant component) has been removed, and the mean of the new signal is zero. This effectively eliminates the constant part of the signal while preserving its varying components.

The ECG (electrocardiogram) signal time series is processed in three stages. The filtering processing in each stage is targeted at different characteristics and processing requirements of the signal. Notch filtering is crucial for removing periodic interference in ECG signals, median filtering is used to remove sharp noise, and wavelet transform can provide local information in time and frequency. By eliminating baseline drift, ECG signals can be made more stable, which helps to more accurately analyze heart rate variability or calculate other physiological parameters. The three stages of filtering together ensure the quality of ECG signals and the accuracy of analysis. Each stage targets different aspects of signal processing to achieve comprehensive optimization of ECG signals.

Convolution kernel

Different categories of data analysis

In order to perform anomaly detection on ECG time series data, it is necessary to select a convolution kernel suitable for ECG data. For the ECG5000 dataset, five categories of ECG data, Normal, PVC, R-on-T, SP and UB, are analyzed, and kernel density estimation and scatter plots are drawn (Fig. 2). Kernel density estimation (KDE) estimates the probability density of data by placing a kernel function on each data point and then adding up the values of all kernel functions.

Kernel functions are usually symmetric, such as Gaussian functions, and have a bandwidth parameter that controls the width of the kernel. It provides a smooth estimate of the data distribution, which makes it very useful in showing the overall trend of the data.

The figure depicted in Fig. 2 can be utilized for analyzing the correlation among different categories of ECG time series data. A two-dimensional kernel density estimation

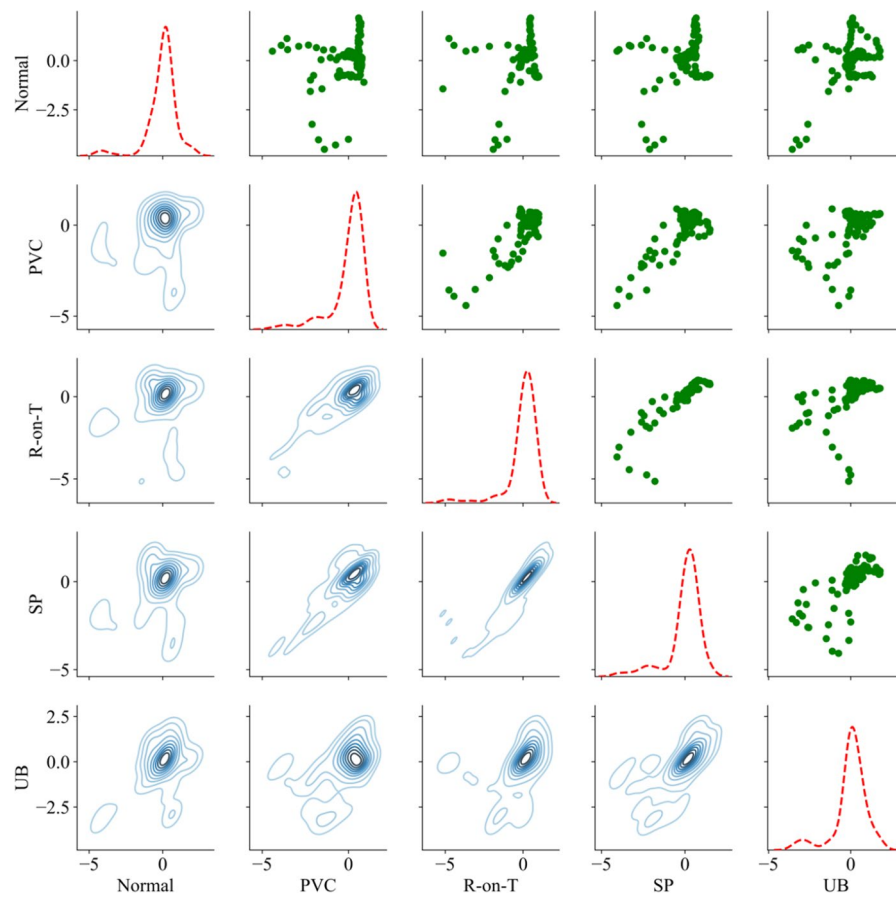


Fig. 2 Statistical analysis of different categories of ECG time series: Two-dimensional kernel density estimation plot (below the diagonal), one-dimensional kernel density estimation plot (diagonal), and scatter plot (above the diagonal)

plot demonstrates the density distributions between two distinct categories of ECG time series. By examining the shape and distribution of the density plot, insights into their correlations and potential category differences can be inferred [31]. There is notably positive correlation between R-on-T and PVC, SP and PVC, SP and R-on-T, R-on-T and UB, SP and UB. All two-dimensional kernel densities reside within a single region, indicating less distinct category differences. This aids in understanding the relationships between different categories, guiding the design of convolutional kernels to capture these specific relationships.

The one-dimensional kernel density estimation graph shows the density distribution of a single ECG time series. The peak and distribution shape of the kernel density estimation graph can be observed to understand the distribution of ECG time series data of different categories. It can be seen that the five categories of ECG time series basically have an obvious peak, which indicates that the data is concentrated near a certain value, which helps to consider this concentration of data when designing the convolution kernel.

The scatter plot shows the relationship between two different types of ECG time series. By observing the distribution pattern of the scatter plot, we can infer the linear

relationship between the two signals, the strength of the correlation, and the possible outliers. Normal is not strongly correlated with the ECG time series of other abnormal categories. Among the four abnormal categories, except for PVC and UB, the distribution of points in the scatter plots of other categories follows a straight line or a path close to a straight line, indicating that the linear relationship is relatively strong. This helps to determine the data correlation that should be paid attention to when designing the convolution kernel.

Tailored convolution kernel

Based on the aforementioned analysis, this study employs an adaptive definition of convolutional kernel matrices for each ECG time series, specifically tailored for Toeplitz convolution. Each kernel's rows represent a time series, encapsulating its mean and standard deviation. Let $X \in R^{n \times s}$ denote the matrix of single-lead ECG time series:

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1s} \\ x_{21} & x_{22} & \cdots & x_{2s} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{ns} \end{pmatrix}_{n \times s} = \begin{pmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_n \end{pmatrix} \quad (5)$$

where, s denotes the length of the sampling time points, while n signifies the number of samples. For each ECG time series $x_i = \{x_{i1}, \dots, x_{is}\}$, compute its mean μ_i , and standard deviation σ_i , and formulate an adaptive convolution kernel K :

$$K = \begin{pmatrix} \mu_1 - 2\sigma_1 & \mu_1 & \mu_1 + 2\sigma_1 \\ \mu_2 - 2\sigma_2 & \mu_2 & \mu_2 + 2\sigma_2 \\ \vdots & \vdots & \cdots \\ \mu_n - 2\sigma_n & \mu_n & \mu_n + 2\sigma_n \end{pmatrix}_{n \times 3} \quad (6)$$

this yields a convolutional kernel matrix K of size $n \times 3$. Such an architecture preserves the primary features of the signal during convolution and adapts dynamically based on the characteristics of different time series to enhance convolutional efficacy.

Toeplitz matrix

For the processing of ECG signal time series, the choice of specific matrix structures depends on the particular applications and requirements. Generally, Toeplitz matrices [32], Hankel matrices [33], and Circulant matrices [34] are commonly employed due to their close relevance to signal processing and ease of handling. Hankel matrices are suited for modeling the stationarity of signals, used in spectrum estimation, signal prediction, and model identification. Circulant matrices are appropriate for modeling periodic signals, useful in discrete Fourier transform (DFT) and circular convolution. The Toeplitz matrix is suitable for simulating linear time-invariant (LTI) systems in the processing of ECG signal time series. First, the Toeplitz matrix can capture the temporal autocorrelation of the signal, which is crucial for analyzing the periodicity and repetitive patterns of ECG signals. Second, the Toeplitz matrix structure is closely related to the convolution operation. Third, since the elements of the Toeplitz matrix are constant on the diagonal, this makes them very efficient in storage and computation. Symmetry

about the main diagonal can be used to optimize the algorithm to speed up the convolution operation. The properties of the Toeplitz matrix make them more intuitive and simple in algorithm implementation.

Specifically, given a matrix, its element T_{ij} at the intersection of the i -th row and j -th column equals a fixed element t_k on some diagonal, where k is determined by $i - j$. Hence, for an $m \times l$ Toeplitz matrix, only $m + l - 1$ parameters are needed for specification. When $m = l$, it takes the form [35]:

$$T = \begin{pmatrix} t_0 & t_{-1} & \cdots & t_{-(m-1)} \\ t_1 & t_0 & \cdots & t_{-(m-2)} \\ & & \ddots & \\ t_{m-1} & t_{m-2} & \cdots & t_0 \end{pmatrix} \quad (7)$$

Constructing a Toeplitz matrix can be achieved through techniques such as kernel convolution, autocorrelation functions, generating functions, or the transpose of Hankel matrices, tailored to different research requirements [36]. This study employs kernel convolution for constructing the Toeplitz matrix T , where the number of rows of T is determined by the data size and the number of rows in kernel K , and the number of columns is determined by the length of the time series samples and the number of columns in K . For instance, if the input data X is of size 4×6 , then according to Eq. (4), a 4×3 convolution kernel K is obtained. Consequently, we define the number of T as $4 + \text{spanclass} = \text{convertEndash}' > 4 - 1 < /span > = 7$ and the number of columns as $6 + \text{spanclass} = \text{convertEndash}' > 3 - 1 < /span > = 8$. To compute each data point in T , for each data x_{ij} in X , we update T_{ij} :

$$T_{ij} = \begin{pmatrix} t_{i-1,j-1} & t_{i-1,j} & t_{i-1,j+1} \\ \vdots & \vdots & \vdots \\ t_{i+n-2,j-1} & t_{i+n-2,j} & t_{i+n-2,j+1} \end{pmatrix} + x_{ij} \bullet (k_{i1}, k_{i2}, k_{i3}) \quad (8)$$

The symbol T_{ij} denotes the data from rows i to $i + n - 1$ of matrix T , and columns j to $j + 2$. $k_i = (k_{i1}, k_{i2}, k_{i3})$ represents the i -th row of the convolution kernel. By cyclically iterating over all i and j , we obtain a Toeplitz matrix $T = (t_{ml}) \in R^{(2n-1) \times (s+2)}$ based on an adaptive convolution kernel.

Fig. 3 illustrates the process of computing the Toeplitz matrix by performing convolution operations on the given data matrix X and kernel matrix.

Adaptive Toeplitz convolution

In the realm of time series (discrete sequences), convolution can be represented as the dot product of two sequences, transformed into the frequency domain via DFT [37]. The convolution operation simplifies to straightforward multiplication in the frequency domain, aligning with the fundamental principle of the Discrete Convolution Theorem [38]. Similarly, convolution can also be achieved through Toeplitz matrices, where the Toeplitz matrix structure correlates with the convolutional kernel matrix of the input signal, thus termed adaptive Toeplitz convolution. In essence, these two types of convolutions are analogous to some extent, both leveraging specific matrix structures to accelerate convolution calculations.

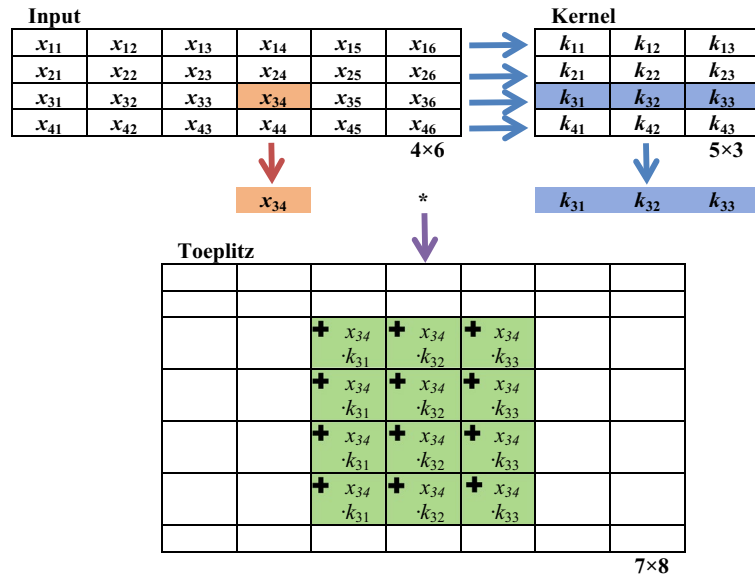


Fig. 3 Construction of Toeplitz matrix

However, the core concept of adaptive Toeplitz convolution lies in dynamically generating convolutional kernels based on the characteristics of input data, dynamically updating the Toeplitz matrix during convolution to adapt to input data features, thereby enhancing model performance and generalization capabilities. Generating convolution based on Toeplitz matrices, $C = (c_{ij}) \in R^{n \times s}$, matches the size of the input ECG time series:

$$C = (c_{ij})_{n \times s} = \begin{pmatrix} \sum_{m=1, l=1}^{n, 3} t_{ml} & \sum_{m=1, l=2}^{n, 4} t_{ml} & \cdots & \sum_{m=1, l=s}^{n, s+2} t_{ml} \\ \vdots & \vdots & & \vdots \\ \sum_{m=n, l=1}^{2n-1, 3} t_{ml} & \sum_{m=n, l=2}^{2n-1, 4} t_{ml} & \cdots & \sum_{m=n, l=s}^{2n-1, s+2} t_{ml} \end{pmatrix} \quad (9)$$

Due to the convolution matrices being the same size as the original data, higher dimensions of the ECG time series do not necessarily enhance the classifier's detection efficiency. Therefore, employing pooling methods to reduce the dimensionality of Toeplitz convolution results while preserving the Toeplitz matrix structure accelerates model computation efficiency and reduces complexity. Here, average pooling [39] is chosen (though maximum pooling could also be considered):

$$p_{i,j//2} = \frac{c_{i,j} + c_{i,j+1}}{2} \quad (10)$$

Due to each stride of 2 in every iteration, where stride defines the increment by which the pooling window moves across input data, $j//2$ denotes a floor division operation, yielding the quotient of j divided by 2 rounded down to the nearest integer. Here, the stride can be any integer greater than 2. In essence, pooling on the signal entails averaging adjacent pairs of elements each time, storing the computed results in a new two-dimensional array for subsequent feature extraction or further processing purposes. The overall structural framework is as depicted in Fig. 4.

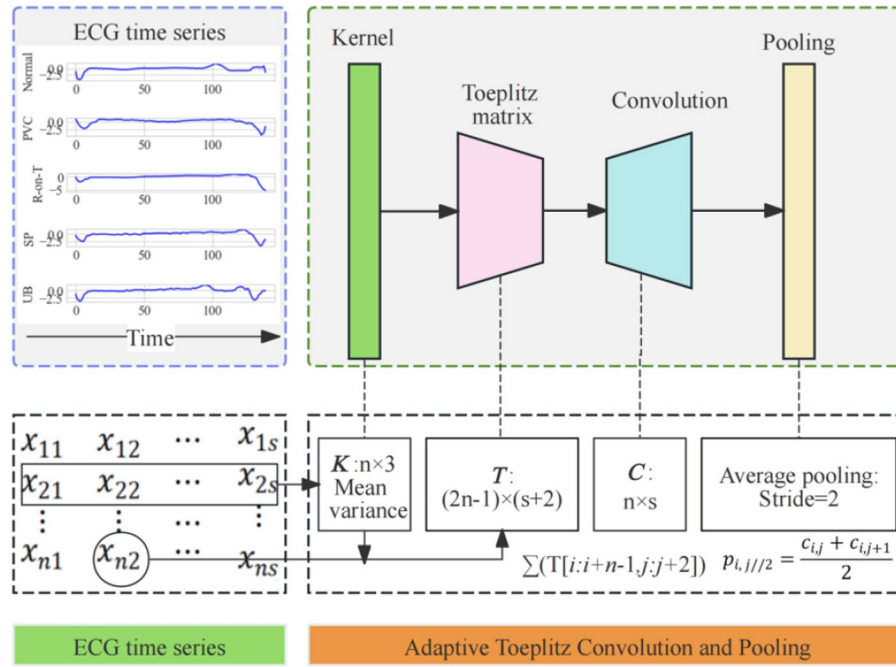


Fig. 4 Construction of Toeplitz convolution and pooling

Proposed model

In this paper, we explore the ECG-ATC model (Fig. 5), which proposes to enhance anomaly detection by transforming ECG time series data through Toeplitz convolution. Specifically, we compare eight different types of classifiers, including tree-based, instance-based, probabilistic models, and linear classifiers (Table 1), to demonstrate the transformation into more structured and correlated features.

Firstly, we preprocess the ECG time series data and then apply Toeplitz convolution, a method that extracts structured features from the time series, enhancing the data's correlation and aiding in subsequent classification tasks. Secondly, to fairly compare the performance of various classifiers in anomaly detection tasks, we take the following steps: we randomly divide the dataset into training and testing sets using the same random seed, ensuring each classifier is trained and tested on the same data subset. Furthermore, we adjust the ratio of the testing set and retrain each classifier. We will then tabulate the anomaly detection results of each classifier under different testing set ratios to evaluate their robustness and effectiveness under different data distributions and sample sizes.

Normal and abnormal data prediction

In order to explore the effect of FDL multi-level signal enhancement preprocessing and adaptive Toeplitz convolution on ECG time series anomaly detection, experiments were conducted on the ECG5000 dataset. The original ECG data, preprocessed ECG data, and convolution pooled ECG data were used for training and testing, and their performance differences were compared. The classifiers were retrained and tested at different test set ratios to verify the stability and generalization ability of the

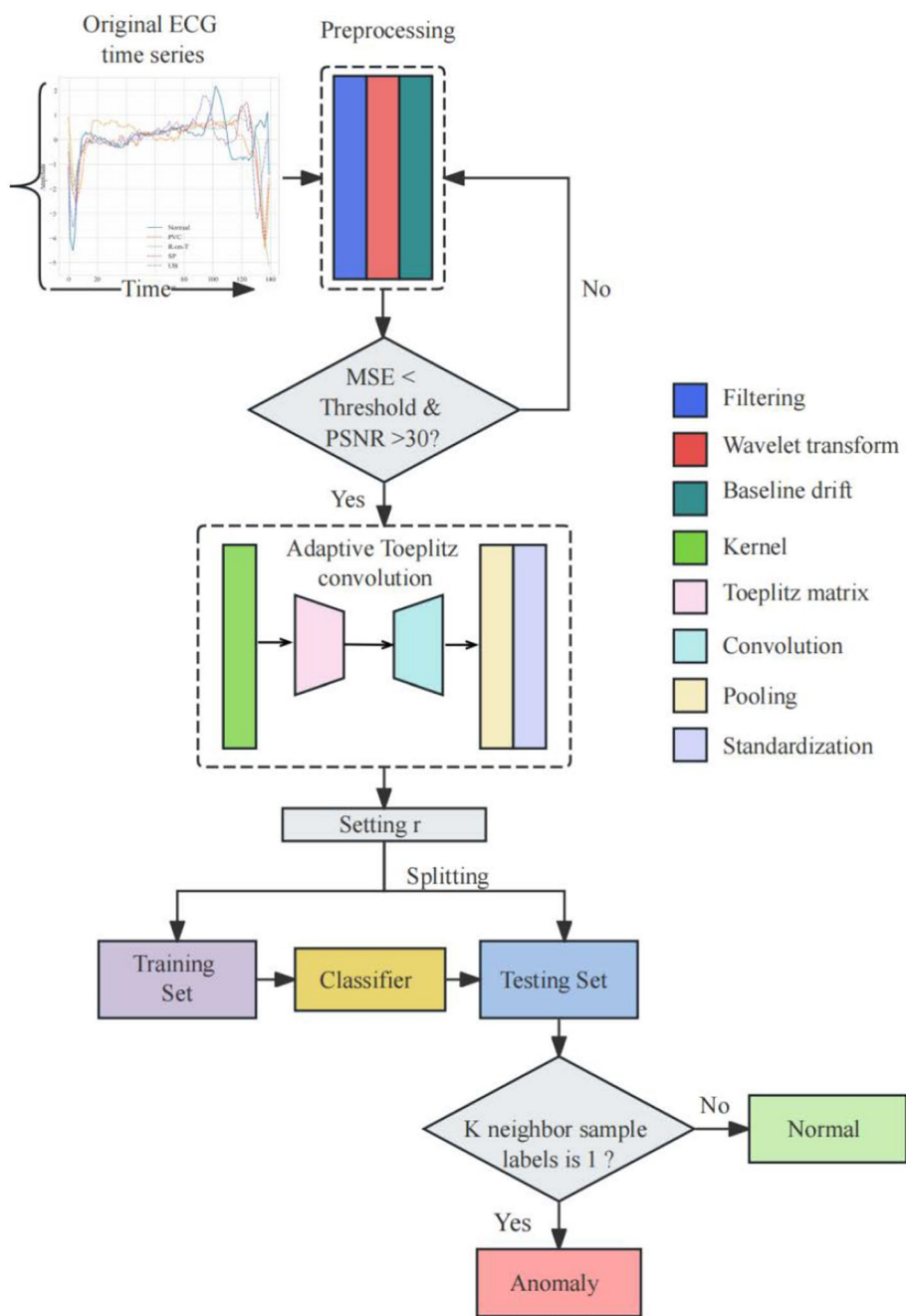


Fig. 5 The framework of the proposed ECG-ATC model

model. Finally, the detection results of each classifier on four types of abnormal ECG data under different conditions were analyzed to evaluate the effectiveness and applicability of the proposed method. The following is a detailed algorithm flow.

Table 1 Utilized models for case study [40]

Classifier	Abbreviation	Family
Random Forest	RF	Ensemble Classifiers
Extreme Gradient Boosting	XGBoost	Ensemble Classifiers
Decision Tree	DT	Decision Tree
K-Nearest Neighbors	KNN	Instance-Based Classifiers
Logistic Regression	LR	Linear Classifiers
Support Vector Machine	SVM	Linear Classifiers
Naive Bayes	NB	Bayesian classifiers
Gaussian Discriminant Analysis	GDA	Bayesian classifiers

Algorithm 1 Proposed ECG-ATC

Input: Data set $X = [x_1, \dots, x_i, \dots, x_n]'$.

Output: Predict whether the test ECG data is normal or anomaly.

Hyper-Parameters: Length of the numerator and denominator coefficient: 3, window size: 3, Sampling frequency: 1000Hz, MSE, Average pooling stride: 2, Random Seed: 42, Test set ratio: r .

```

1: Begin
2:   Preprocessing the ECG time series:  $X_{prep}$ .
3:   Initialize the Toeplitz matrix:  $T = [0]_{(2n-2) \times (s+2)}$ .
4:   Calculate the mean  $\mu_i$  and variance  $\sigma_i$  of  $x_i$ .
5:   Toeplitz matrix:  $T[i: i + n - 1, j: j + 2] \oplus = x_{ij} * (\mu_i - 2\sigma_i, \mu_i, \mu_i + 2\sigma_i)$ .
6:   Convolution matrix:  $c_{ij} = \sum_{m=i, l=j}^{i+n-1, j+2} t_{ml}$ .
7:   Pooling matrix:  $p_{i,j/2} = 0.5 * (c_{i,j} + c_{i,j+1})$ .
8:   Convolution pooling of ECG time series:  $X_{pool}$ .
9:    $r = 0.1$ , split the training set and test set for  $X, X_{prep}, X_{pool}$ .
10:  /*Training Phase*/
11:  For 8 classifiers do
12:    Initialization parameters.
13:    Training the classifier.
14:  end for
15:  /*Testing Phase*/
16:  for 8 classifiers do
17:    Calculate the anomaly score of the test data to classify normal and abnormal ECG signals.
18:  end for
19:  for  $r$  in range (0.05, 0.9) and interval 0.05, do
20:    Split the training set and test set for  $X_{pool}$ .
21:    Comparison of the stability of the proposed model.
22:  end for
23:  for  $i$  in range (1, 4) do
24:    Calculate the detection results of the proposed model in the  $i$ -th anomaly category.
25:  end for

```

Experimental evaluation

The aim of this study is to evaluate the proposed ECG-ATC model in detecting anomalies in ECG signals, focusing primarily on enhancing efficiency and accuracy. To validate the effectiveness of the proposed FDL multi-stage signal enhancement approach, it is essential to compare the quality of ECG signal preprocessing and assess how the

preprocessed ECG time series impacts the accuracy and efficiency of anomaly detection. Furthermore, to assess the efficacy of adaptive Toeplitz convolution in anomaly detection of ECG time series, different classifiers will be employed to compare the results of anomaly detection on convolved and pooled ECG signals.

Evaluation of preprocessing results

The preprocessing outcome aims to extract key features of ECG time series while preserving similarity to the original data, measured by Mean Squared Error (MSE) and Pearson correlation coefficient. A lower MSE indicates better reconstruction. Additionally, Peak Signal-to-Noise Ratio (PSNR) serves as a quality metric, typically used to assess compression algorithm performance [41]. Higher PSNR values denote better quality. The computation of these metrics is as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n \|x_i - y_i\| \quad (11)$$

where, x_i and y_i respectively denote the values before and after signal processing, with n representing the number of samples in the ECG time series. The correlation [42]:

$$Corr = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (12)$$

where $\bar{x}$ and $\bar{y}$ correspond to the mean values of the time series before and after processing, the correlation here ranges from -1 to 1 . A correlation of -1 indicates complete negative correlation, while 1 indicates complete positive correlation, and 0 indicates no correlation.

The PSNR value is obtained by calculating the ratio between the signal peak and the noise variance, with the formula as follows:

$$PSNR = 10 * \log_{10} \frac{p^2}{MSE} \quad (13)$$

where p represents the maximum possible value of the signal, indicating the degree of difference between the original signal and the compressed signal. Typically, a PSNR value of 30 dB or higher is considered indicative of high quality, while values between 20–30 dB suggest moderate quality, and below 20 dB are deemed lower quality.

Evaluation of anomaly detection results

The anomaly detection of ECG time series is framed as a binary classification problem. Table 2 presents the confusion matrix, where FP denotes the occurrence of false

Table 2 Confusion Matrix For binary classification

Actual\Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

positives when normal instances are incorrectly classified as anomalies, and FN signifies misclassification of anomaly instances as normal. Similarly, TP and TN respectively denote accurate identification of anomaly and normal instances.

Five distinct evaluation metrics are predominantly employed: AUC, Accuracy, Precision, Recall, F1 Score, and execution time (S). Then, different metrics can be calculated as follows [43]:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

$$Precision = \frac{TP}{TP + FP} \quad (15)$$

$$Recall = \frac{TP}{TP + FN} \quad (16)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (17)$$

Experimental observations on data efficiency of ECG-ATC methods

Experimental setup

The present study proposes a model for anomaly detection in ECG time series data. It involves preprocessing, time series transformation, training, and testing phases. All experiments were conducted using Python, utilizing the signal module from SciPy for signal processing and filtering operations, enabling the design of various filters, spectral computations, and Fourier transforms. Wavelet transformation employed the fourth Daubechies wavelet, 'db4', with a threshold of 0.05. Data standardization was performed to achieve a distribution with mean 0 and variance 1, influencing data distribution and inter-point distances, thereby impacting feature learning and classification outcomes. Various classifier models were loaded using the sklearn machine learning library.

Overall hyperparameters of the method were empirically determined, with default parameters used for each classifier across all experiments in Python, maintaining consistency across different test and training datasets. Here, we varied the ratio (r) between different test and training sets, altering the size of the training set to monitor model performance on the test set. To compare the effectiveness of preprocessing and Toeplitz convolution for anomaly detection in ECG time series, an initial r of 0.9 was chosen, with Table 3 summarizing the empirically determined hyperparameters used in the proposed framework.

Preprocessing results

To standardize the baseline of ECG time-series data, a multi-stage signal enhancement process combining baseline drift removal with composite filtering and wavelet denoising is employed. This approach leverages their respective strengths while compensating for their individual limitations, thus comprehensively addressing noise, interference, and

Table 3 Overview of ECG-ATC model hyper parameters

Parameters	Value
Length of the numerator coefficient (M)	3
Length of the denominator coefficient (N)	3
window size ($2k + 1$)	3
Sampling frequency	1000 Hz
Loss	MSE Loss
Average pooling stride	2
Random seed	42
Test set ratio (r)	0.9

Table 4 The metrics of ECG time series enhanced by FDL

Categories	MSE	Correlation	PSNR
Normal	0.029	0.986	31.364
PVC	0.015	0.994	31.60
R-on-T	0.007	0.998	36.55
SP	0.008	0.997	35.42
UB	0.012	0.996	32.79

baseline drift issues within the signal. Table 4 provides comprehensive processing results based on MSE, correlation, and PSNR values.

Table 4 illustrates that employing the proposed FDL multi-stage enhanced preprocessing technique keeps the MSE of the 5 classes of ECG signals in the ECG5000 dataset under 0.03. The correlation with the original signal post-processing of normal class ECG time series is 0.986, while the remaining 4 abnormal categories exceed 0.99. PSNR values are all greater than 30, indicating the proposed preprocessing method incurs low loss and maintains high quality.

Fig. 6 presents a comparison between the original and reconstructed signals post preprocessing for ECG signals from normal, PVC, R-on-T PVC, SP, and UB categories. The left side displays the time series curves, while the right side features scatter plots with correlation lines. This visually demonstrates the efficacy of the reconstruction algorithm across different types of ECG signals and the consistency achieved between the reconstructed and original signals.

Experimental results for $r = 0.9$

When the proportion $r = 0.9$ for the test set, the training set comprises 500 samples of both normal and abnormal ECG time series, while the test set consists of 4500 samples, including both normal and abnormal samples. Choosing $r = 0.9$ results in a larger test set compared to the training set, aiming to more prominently highlight differences in detection outcomes. The random seed is set to 42 for the purpose of partitioning the training and test sets, ensuring experiment reproducibility. To demonstrate the effectiveness of the proposed ECG-ATC model for anomaly detection on the ECG5000 dataset, comparisons of anomaly detection results among eight classifiers are conducted on the

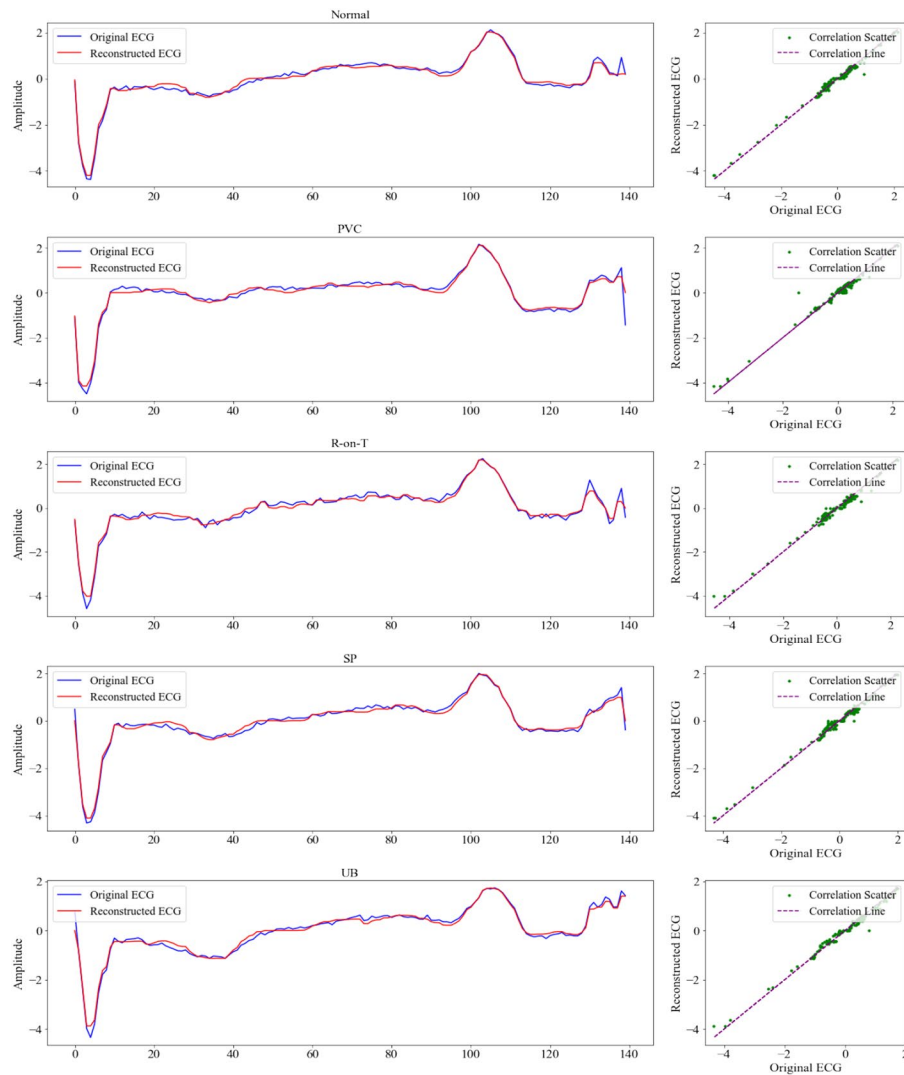


Fig. 6 ECG time series (left) and scatter plots (right) enhanced by FDL for each category

original ECG time series dataset, the preprocessed dataset, and the dataset after convolutional pooling. Fig. 7 visualizes the comparative accuracy measurements.

Figure 7 shows the accuracy performance of different classifiers under three different data processing methods. On the raw ECG time series data without any preprocessing, most classifiers showed high accuracy, among which the SVM classifier performed best, reaching 99%. This shows that without additional feature engineering, these classifiers are able to capture the characteristics of ECG data well. The test accuracy of the preprocessed data on all classifiers is almost unchanged compared with the raw data, except that the accuracy of DT decreased slightly (0.7%). XGBoost performed slightly better (0.5%) on the preprocessed data. This means that the preprocessing steps have limited impact on the performance of these classifiers. The data after Toeplitz convolution and pooling operations showed improved accuracy on some classifiers, especially KNN, GDA, DT, XGBoost, and RF, with improvements of 1.2%, 1.6%, 2.8%, 2.0%, and 0.9%, respectively.

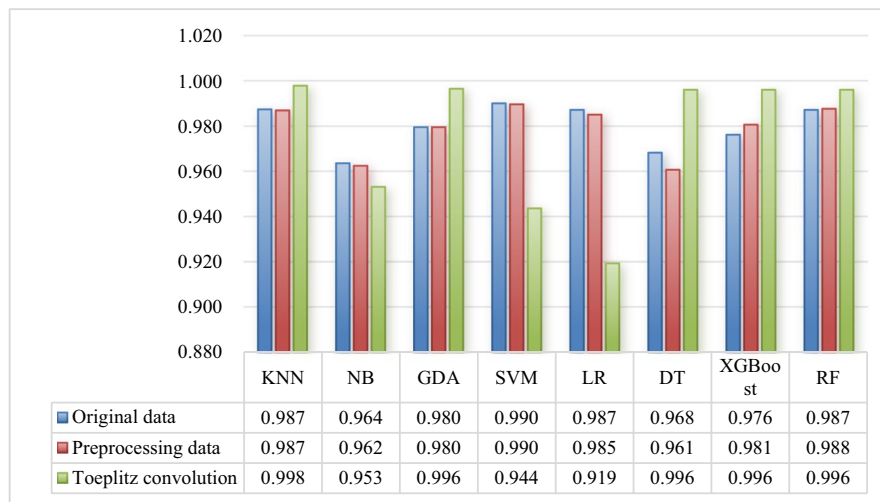


Fig. 7 Accuracy on the ECG5000 test set

This suggests that Toeplitz convolution may help these algorithms better identify patterns in the data.

However, for NB, SVM, and LR, the accuracy of the data processed by Toeplitz convolution decreases by 0.9%, 4.6%, and 6.6%, respectively. This is because Toeplitz convolution increases the nonlinearity of features and is not suitable for linear classifiers such as LR and SVM.

Table 5 presents additional detection metrics including AUC, precision, recall, and F1 score, providing a comprehensive evaluation of the impacts of different preprocessing techniques. Table 5 indicates that KNN, GDA, DT, XGBoost, and RF perform best in terms of AUC, precision, recall, and F1 score on the Toeplitz convolution dataset, with DT's F1 score 3.3% higher than the original data. Furthermore, the performance of these 5 classifiers on the preprocessed dataset is the same as or slightly better than that on the original data, with XGBoost's precision 1.1% higher than the original data, only slightly lower in the preprocessed results for DT. NB, SVM, and LR have the highest AUC values in the original data, consistent with the accuracy performance in Fig. 7, but the precision and recall are highest in the Toeplitz convolution data, even reaching 1, meaning they can identify all abnormal ECG time series samples, which is meaningful in the medical field.

Comprehensive evaluation shows that for the task of ECG data anomaly detection, considering that the original ECG data usually has problems such as long sampling time, high dimensionality, and noise interference, the proposed ECG-ATC method aims to extract important dependencies in the time series while reducing the data dimension by processing the ECG time series. This method may help improve the accuracy and efficiency of detection, and has a certain robustness to the interaction between features. However, this method is not suitable for all types of classifiers. Toeplitz convolution can enhance the performance of the algorithm in some cases, especially in those algorithms that can exploit complex feature relationships. However, for linear classifiers such as logistic regression and support vector machines, Toeplitz convolution may not be the best choice because it may introduce additional

Table 5 Detection metrics of 8 classifiers on ECG time series with different processing

Model	Metrics	Original data	Preprocessing data	Toeplitz convolution
KNN	AUC	0.986	0.985	0.998
	Precision	0.993	0.993	0.998
	Recall	0.995	0.995	0.999
	F1	0.985	0.984	0.997
NB	AUC	0.967	0.966	0.945
	Precision	0.929	0.926	0.991
	Recall	0.946	0.943	0.994
	F1	0.958	0.956	0.941
GDA	AUC	0.980	0.979	0.996
	Precision	0.969	0.973	0.998
	Recall	0.978	0.981	0.999
	F1	0.976	0.976	0.996
SVM	AUC	0.990	0.990	0.932
	Precision	0.987	0.986	1
	Recall	0.990	0.990	1
	F1	0.988	0.987	0.927
LR	AUC	0.988	0.985	0.903
	Precision	0.979	0.977	1
	Recall	0.985	0.984	1
	F1	0.985	0.982	0.893
DT	AUC	0.968	0.959	0.996
	Precision	0.959	0.955	0.995
	Recall	0.970	0.968	0.996
	F1	0.962	0.953	0.995
XGBoost	AUC	0.977	0.980	0.996
	Precision	0.964	0.975	0.995
	Recall	0.974	0.982	0.996
	F1	0.972	0.977	0.995
RF	AUC	0.987	0.987	0.996
	Precision	0.983	0.985	0.995
	Recall	0.988	0.989	0.996
	F1	0.985	0.985	0.995

The best results are highlighted in bold

nonlinear features that are difficult for linear models to effectively handle. In addition, for the naive Bayes classifier, its performance relies on the independence assumption of the features, and the correlation of ECG time series data may conflict with this assumption.

Figure 8 presents the detection time of various classifiers on three test ECG time series datasets through line charts. Due to the limited training set of only 500 ECG samples, the overall training time is relatively short. Preprocessed data exhibits consistent runtime across most classifiers, namely GDA, SVM, and XGBoost, with slightly lower times compared to raw data. The ECG time series, reduced in dimension by half through Toeplitz convolution and pooling operations, shows varying runtimes among models. SVM notably exceeds both raw and preprocessed data in

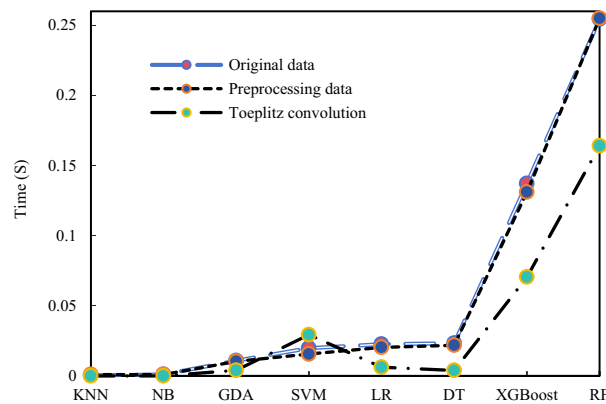


Fig. 8 Run time of the test set ECG time series

runtime, while the other seven models exhibit the lowest times, with the largest disparity observed between XGBoost and RF.

In conclusion, ECG time series data processed through Toeplitz convolution and pooling operations are more suitable for anomaly detection using tree-based ensemble algorithms, GDA, and KNN. The chosen average pooling with a stride of 2 aligns with the original length of 140 for the ECG5000 dataset, reduced to 70 after convolution and pooling. For ECG time series with more sampling points, longer strides in average or maximum pooling can further enhance dimensionality reduction.

The subsequent sections primarily utilize KNN, GDA, DT, XGBoost, and RF classifiers to experimentally demonstrate the generalization and robustness of the proposed ECG-ATC model.

Experimental results for different r values

In Section Experimental Results for $r = 0.9$, a test set ratio $r = 0.9$ was employed only. To further evaluate the generalizability and robustness of the proposed ECG-ATC model, different test set ratios were configured. Fig. 9 depict the variations in the model's AUC (%) and accuracy (%) on testing ECG signals as the test set ratio varies.

Figure 9 illustrates the comparison among three different approaches: the original dataset, the preprocessed dataset, and the Toeplitz convolution processed dataset, as the proportion r of the test ECG time series dataset increases gradually from 0.05 to 0.9. It evaluates the performance of five classifiers in terms of AUC, Accuracy, F1 score, and runtime.

Figure 9a and b depict the performance of the five classifiers across 18 different r values (ranging from 0.05 to 0.9) in terms of AUC and Accuracy using box plots. It is evident that on the Toeplitz convolution processed dataset, the AUC and Accuracy values of all five classifiers are significantly higher compared to those on the original and preprocessed datasets, consistently exceeding 99.5%. Regardless of variations in the test set proportion (r value), both AUC and accuracy remain highly concentrated, indicating strong generalization of the classifiers on the Toeplitz convolution processed dataset and stable classification results.

Figure 9c and d present the variation trends of F1 score and runtime of the five classifiers under different dataset processing methods using grouped error line graphs.

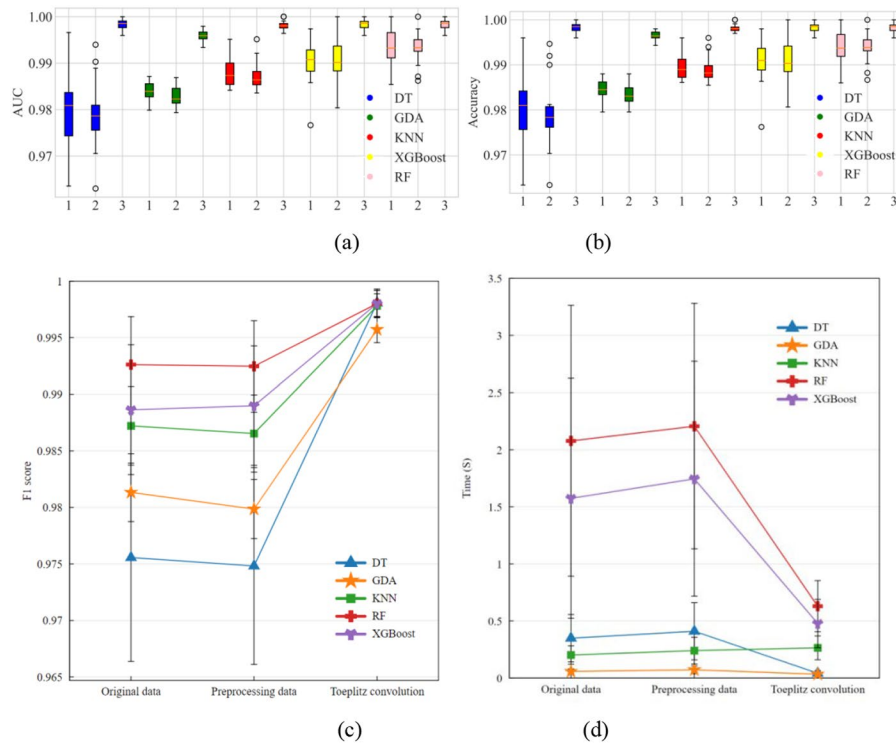


Fig. 9 Results for r ranged from 0.05 to 0.9. **a** Grouped box plot of AUC; **b** Grouped box plot of Accuracy; 1: Original data, 2: Preprocessing data, 3: Toeplitz convolution. **c** Line chart of group error of F1 score; **d** Line chart of group error of run time

Across various r values, the F1 score performance of the classifiers on the Toeplitz convolution processed dataset remains the most stable and consistently highest. Compared to the original dataset, the F1 score on the Toeplitz convolution processed dataset has improved by approximately 2%, further highlighting its superiority.

Apart from the KNN classifier, the proposed models exhibit the shortest runtime on the Toeplitz convolution processed dataset. This significant time advantage demonstrates that Toeplitz convolution processing not only enhances classification performance but also exhibits clear efficiency in runtime. In summary, Fig. 9 clearly demonstrates that the proposed Toeplitz convolution processing method exhibits strong generalization and stability when handling ECG time series datasets. Regardless of variations in the test dataset size, the classifiers perform outstandingly in terms of AUC, Accuracy, and F1 score on the Toeplitz convolution dataset, with significantly reduced runtime. This indicates the method's effectiveness in addressing classification tasks across diverse testing scenarios, making it a superior data processing technique.

Results for each category

After implementing Toeplitz convolution processing, each abnormal class's ECG time series, along with all normal class data, were trained and tested using KNN, GDA, DT, RF, and XGBoost. The parameter $r = 0.2$ was set, and Fig. 10 illustrates the confusion matrices of various classifiers on the test set [44].



Fig. 10 Confusion matrix for each category detection. **a** KNN; **b** GDA; **c** DT; **d** RF; **e** XGBoost

Figure 10 utilized the confusion matrix to evaluate the recognition effect of five classifiers on four types of abnormalities in the ECG time series: PVC, R-on-T, SP, and UB. The experimental results show that all classifiers have achieved perfect detection accuracy in the UB and SP categories, and in the R-on-T category, except for GDA, the other classifiers also show extremely high accuracy. Although GDA has a small number of misclassifications in the PVC category, overall, the detection accuracy of all classifiers in each category is above 99.6%, which verifies the effectiveness of the proposed ECG-ATC model in processing ECG time series data.

Comparison between related work and proposed model

To validate the efficacy of our proposed algorithm, it is imperative to compare it with other existing algorithms that share similar properties and utilize the same dataset. Conducted on the ECG5000 dataset, the task involves binary classification between normal and abnormal classes. This section addresses the comparison of the proposed ECG-ATC model with state-of-the-art approaches, focusing on evaluating the accuracy of abnormal ECG signal detection across different training and testing set partitions. Table 6 encapsulates the comparison between the classification accuracy of the proposed ECG-ATC model and the reported results from relevant existing literature [46–50, 23], thereby affirming the efficacy of the proposed approach. The pinnacle accuracy among the existing literature is attributed to Biloborodova et al. [47], who amalgamated linear and non-linear electrocardiogram dynamics for feature analysis, conducting experiments on the ECG5000 dataset with 500 Normal samples and 500 R-on-T premature ventricular contraction abnormal samples, achieving a remarkable accuracy of 99.5%. From the summary, it can be seen that the proposed ECG-ATC method using FDL pre-processing and adaptive Toeplitz convolution technique achieved the highest accuracy among the compared methods.

Experimental results for PTB

To further verify the generalization ability of the proposed model, the PTB Diagnostic ECG Database is a dataset widely used in ECG research. Different from the ECG5000

Table 6 Performance comparison of ECG-ATC model with other existing methods in terms of accuracy

Author	Model	<i>r</i>	Accuracy (%)
Dosovitskiy et al. [45]	ViT	0.2	97.67
		0.9	95.12
Dutta et al. [46]	LSTM Auto-encoder	0.9	97.93
Biloborodova et al. [47]	PQRST complexes attributes	-	99.5
Roy et al. [22]	ECG-NET with manual threshold	0.533	98.68
	ECG-NET with automated threshold	0.533	98.42
	ECG-NET with manual threshold	0.9	98.56
	ECG-NET with automated threshold	0.9	98.36
Alamr and Artoli [48]	Unsupervised Transformer	0.1	99
Panchal et al. [49]	Multimodal Image Fusion (MIF)	–	97.79
	Multimodal Feature Fusion (MFF)	–	98.19
This study	ECG-ATC with KNN	0.1	100.00
		0.9	99.78
	ECG-ATC with GDA	0.1	99.80
		0.9	99.64
	ECG-ATC with DT	0.1	100.00
		0.9	99.60
	ECG-ATC with XGBoost	0.1	100.00
		0.9	99.60
	ECG-ATC with RF	0.1	100.00
		0.9	99.60

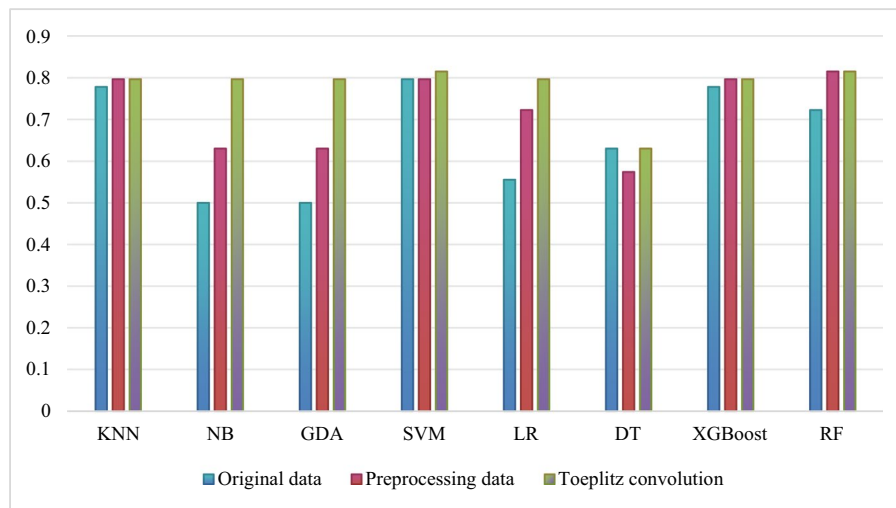


Fig. 11 Accuracy on the PTB test set

dataset, it contains 12-lead signals from 268 patients. The detection accuracy of the original data, preprocessed data, and Toeplitz convolution data is also compared.

Figure 11 shows that for KNN, XGBoost, and RF models, Toeplitz convolution processing improves the original data, but does not bring significant performance improvement compared to preprocessed data. In contrast, NB, GDA, and LR models show significant performance improvement after Toeplitz convolution processing. The SVM model also shows a slight performance improvement after Toeplitz convolution processing, with the accuracy increasing from 0.8 to 0.815. The performance of the DT model decreases after preprocessing, but is the same as the original data after Toeplitz convolution processing. The prediction accuracy of PTB did not reach the level of the ECG5000 dataset, which is related to the small amount of data in the PTB database, affecting the adequacy of model training and generalization ability. These results show that Toeplitz convolution processing can effectively extract key features in ECG signals for some models, especially NB and GDA, thereby improving classification accuracy. However, this processing method is not suitable for all types of classifiers, especially for linear classifiers, and its performance improvement may be limited. These findings provide valuable insights into anomaly detection in ECG signals and further emphasize the need to consider the characteristics of the chosen classifier before applying Toeplitz convolution.

Conclusion

The manuscript presents a novel approach that integrates a multi-stage enhancement preprocessing method for ECG time series data, utilizing FDL, with an adaptive Toeplitz convolutional model for the detection of arrhythmias in ECG time series data. Initially, the ECG signals are preprocessed through multiple stages leveraging FDL, followed by the integration of adaptive Toeplitz convolutional operations, which effectively learn the patterns of both normal and abnormal heartbeats, thereby significantly enhancing classification accuracy. Furthermore, to improve detection efficiency, the data post-convolution is subjected to an average pooling process. On

the publicly available ECG5000 dataset, the proposed model was extensively evaluated and its performance was compared with other state-of-the-art methods. Experimental results show that the proposed model outperforms the comparative methods in the literature, achieving an accuracy of 99.6%. It is further validated on the PTB dataset. However, there are still some limitations and potential room for improvement. Multiple parameters in the model (number of stages of FDL preprocessing) need to be carefully tuned to achieve optimal performance. The performance of the model on the ECG5000 dataset may not be fully generalized to other datasets, especially those with different characteristics or noise levels. For linear classifiers such as logistic regression and support vector machines, Toeplitz convolution may not be the best choice because it may introduce nonlinear features that are difficult to handle with linear models. Combining the ECG-ATC model with other physiological signals (such as blood pressure, respiratory rate) to build a multimodal heart health monitoring system may bring challenges in data fusion and feature selection. Future research endeavors to further optimize the adaptive Toeplitz convolution and FDL preprocessing methods in order to enhance computational efficiency and reduce model complexity. Explorations into integrating the proposed ECG-ATC model with other physiological signals, such as blood pressure and respiratory rate, to construct a multimodal cardiovascular health monitoring system, are also envisioned. Additionally, the development of corresponding real-time detection systems that utilize the proposed model for real-time ECG monitoring is proposed, thereby augmenting the clinical application value.

Abbreviations

ECG	Electrocardiogram
AE	Autoencoder
ML	Machine learning
AUC	Area Under the Receiver Operating Characteristic Curve
ECG-ATC	Adaptive Toeplitz convolution ECG time series model
PVC	Premature Ventricular Contraction
IIR	Infinite Impulse Response
R-on-T	R-on-T Premature Ventricular Contraction
SP	Supraventricular Premature or Ectopic Beat
UB	Unclassified Beat
FDL	Composite filtering, denoising, and baseline removal
RF	Random Forest
XGBoost	Extreme Gradient Boosting
DT	Decision Tree
KNN	K-Nearest Neighbors
LR	Logistic Regression
SVM	Support Vector Machine
NB	Naive Bayes
GDA	Gaussian Discriminant Analysis
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
r	Ratio between different test and training sets

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Author contributions

LW and MA provided the concept; LW and TL provided the methodology; CN provided software support; LW, MA, YT, and XZ provided supervision; LW, MA, and TL performed partial validation; YT and XZ performed the visualization and made Fig. 9; LW wrote the manuscript. All authors have read and approved the published version of the manuscript.

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Availability of data and materials

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This article does not contain any studies with human participants or animals performed by any authors.

Consent for publication

I hereby give my consent to publish the research article. I confirm that all authors have agreed to the publication. I understand that the article will be made publicly available.

Competing interests

The authors declare no competing interests.

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